

Network Traffic Analysis

Wireshark & TCP/IP - Technical Report

Scope	ARP, DNS, TCP, HTTP, ICMP
Capture	15 synthetic packets
Client	192.168.56.10
Server	203.0.113.80
Domain	training.example

1. Objective

The purpose of this lab was to inspect a controlled packet capture in Wireshark and identify how common network protocols appear on the wire. The analysis focuses on ARP, DNS, TCP, HTTP, and ICMP, with particular attention to address resolution, name resolution, TCP connection establishment, application-layer requests, and connection teardown.

2. Lab Environment

The capture is synthetic and uses reserved lab addressing. The client is **192.168.56.10**, the DNS server is **192.168.56.53**, the gateway is **192.168.56.1**, and the web server is **203.0.113.80**. The hostname is **training.example**. This makes the capture suitable for a public portfolio without exposing real production traffic.

3. Capture Overview

Packets	Protocol	Observed activity
1-2	ARP	Gateway MAC address resolution
3-4	DNS	A-record query and response
5-7	TCP	Three-way handshake to TCP/80
8-9	HTTP	GET request and 200 OK response
10-13	TCP	Acknowledgment and orderly connection close
14-15	ICMP	Echo request and reply

4. Findings

4.1 ARP - Packets 1-2

Packet 1 is an ARP request from **192.168.56.10** asking which MAC address owns **192.168.56.1**. The Ethernet destination is broadcast because the client does not yet know the gateway hardware address. Packet 2 is the gateway reply. **Interpretation:** ARP maps a local IPv4 address to a Layer 2 MAC address so an Ethernet frame can be delivered to the next hop.

4.2 DNS - Packets 3-4

Packet 3 is a UDP DNS query from client source port **53000** to server destination port **53**, requesting the A record for **training.example**. Packet 4 returns **203.0.113.80**. The client port is ephemeral, while port 53 identifies the DNS service.

4.3 TCP Three-Way Handshake - Packets 5-7

The client opens a TCP connection from ephemeral port **51514** to web-server port **80**. Packet 5 carries SYN, packet 6 carries SYN/ACK, and packet 7 carries ACK. This establishes shared TCP state before application data is transferred.

4.4 HTTP - Packets 8-9

Packet 8 carries the HTTP request below. Packet 9 contains **HTTP/1.1 200 OK** and a small HTML body.

```
GET /lab/index.html HTTP/1.1
Host: training.example
```

Security observation: this lab uses plaintext HTTP, so request headers and content are visible in the capture. HTTPS is designed to encrypt the HTTP application data.

4.5 TCP Teardown - Packets 10-13

After the HTTP response is acknowledged, FIN/ACK packets close each direction of the TCP session. Connection termination is directional, so each endpoint indicates that it has finished sending and acknowledges the peer close.

4.6 ICMP - Packets 14-15

Packet 14 is an ICMP Echo Request and packet 15 is an Echo Reply. This is the basic exchange used by **ping**. A reply shows that these ICMP packets traversed the path, but it does not prove that every application service is available.

5. Wireshark Filters Used

```
arp
dns
tcp
tcp.port == 80
http
icmp
ip.addr == 192.168.56.10
ip.addr == 203.0.113.80
```

To isolate initial TCP connection attempts, use:

```
tcp.flags.syn == 1 && tcp.flags.ack == 0
```

6. TCP/IP Model Mapping

Observed data	TCP/IP layer	Purpose
Ethernet / ARP	Link	Local frame delivery and address resolution
IPv4 / ICMP	Internet	Addressing, routing, diagnostics
TCP / UDP	Transport	Ports, reliable byte streams, or datagrams
DNS / HTTP	Application	Name resolution and web communication

7. Security Takeaways

Packet analysis can reveal which systems communicated, which services and ports were used, how transport sessions behaved, and what plaintext application data crossed the network. However, normal protocol behavior alone is not evidence of malicious activity. Security conclusions require context, baselines, endpoint evidence, timing, and often many more events.

Encrypted protocols change visibility. With HTTPS, metadata such as IP addresses, ports, timing, packet sizes, and parts of the TLS handshake may remain observable, while the HTTP content is normally encrypted.

8. Conclusion

The capture demonstrates the path from local address resolution through DNS, TCP connection setup, an HTTP request/response, connection termination, and ICMP reachability testing. The exercise connects raw packet fields to the TCP/IP model and shows how Wireshark turns individual frames into an explanation of network behavior.

9. Limitations

This is a small controlled capture rather than production traffic. It does not include TLS, IPv6, retransmissions, packet loss, malicious traffic, or high-volume flows. Those are appropriate follow-up labs after the fundamentals are comfortable.